

Advanced Storyboarding & UI Prototyping

Online Hospital Management System — **MediCare+**

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Course	HCI & Computer Graphics Lab (CS341)
Lab	Lab 12 - Advanced Storyboarding & UI Prototyping (CLO-3)
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Tool	Figma (high-fidelity, interactive prototype)
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1. Scenario Description

This project designs a high-fidelity storyboard and interactive prototype for an **Online Hospital Management System** branded **MediCare+**. The system lets patients register, search and filter doctors, book appointments, pay consultation fees online, and receive reminders — replacing slow manual reception queues. The design follows core usability principles (consistency, visibility, feedback, error prevention) and WCAG 2.1 accessibility guidance throughout.

Key features covered

- Patient registration & secure login
- Doctor search with live filters (rating, availability, fee)
- Appointment booking with date / time selection
- Secure online payment with transparent fee breakdown
- Prescription & visit records, reminders and emergency support

2. User Persona

The primary persona is **Sara Ahmed**, a 22-year-old intermediate-skill university student who is busy and goal-driven. Her main goal is to book a doctor quickly; her biggest frustrations are cluttered navigation and a lack of feedback. She has mild red/green colour-blindness, so status is always conveyed with an icon and text, never colour alone.

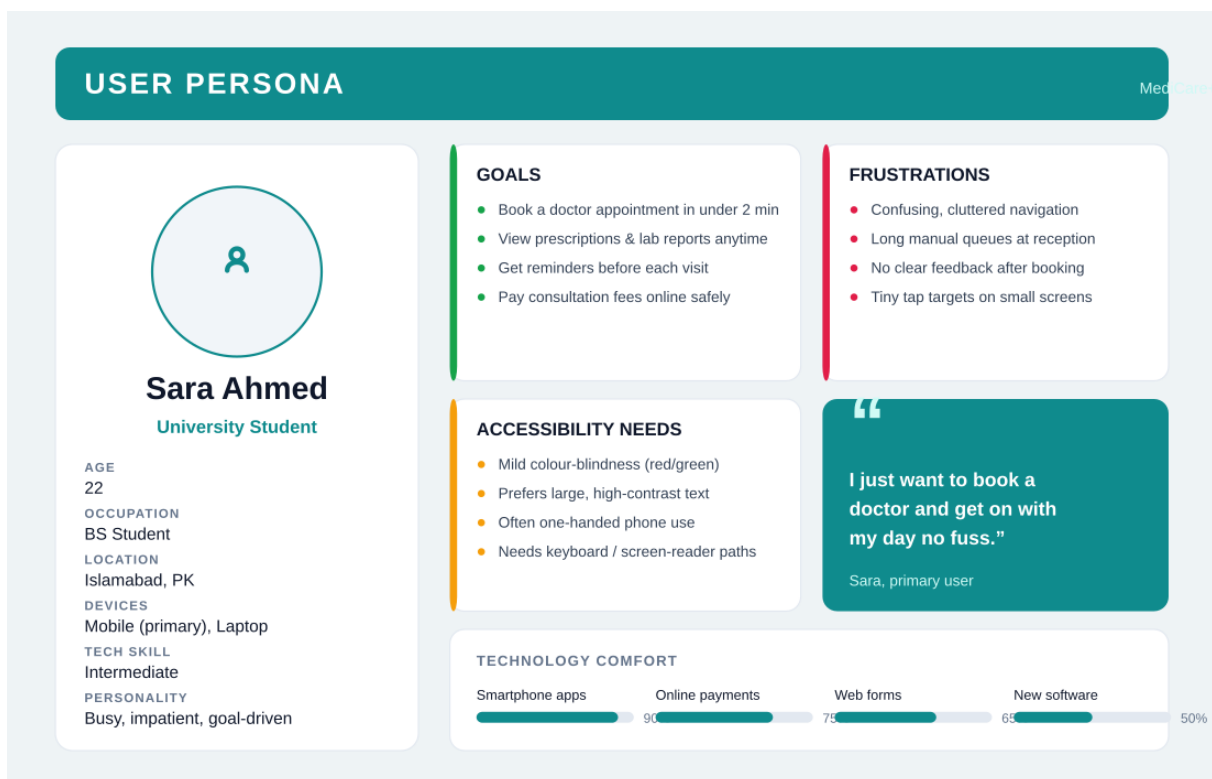


Figure 1 - Detailed user persona: goals, frustrations, accessibility needs and tech comfort.

3. Primary & Secondary Goals

Primary goal: Book a doctor appointment successfully in under two minutes.

Secondary goals:

- View prescription & lab-report history
- Receive appointment reminders
- Pay consultation fees online securely
- Contact / join the doctor online

4. User Journey Flow

The journey spans ten interaction steps from launch to confirmation and deliberately includes a **decision point** (credential validation) and an **error scenario** (wrong password) with a clear recovery path. The emotion curve dips at the login error and peaks at the success screen, showing how good feedback restores user confidence.

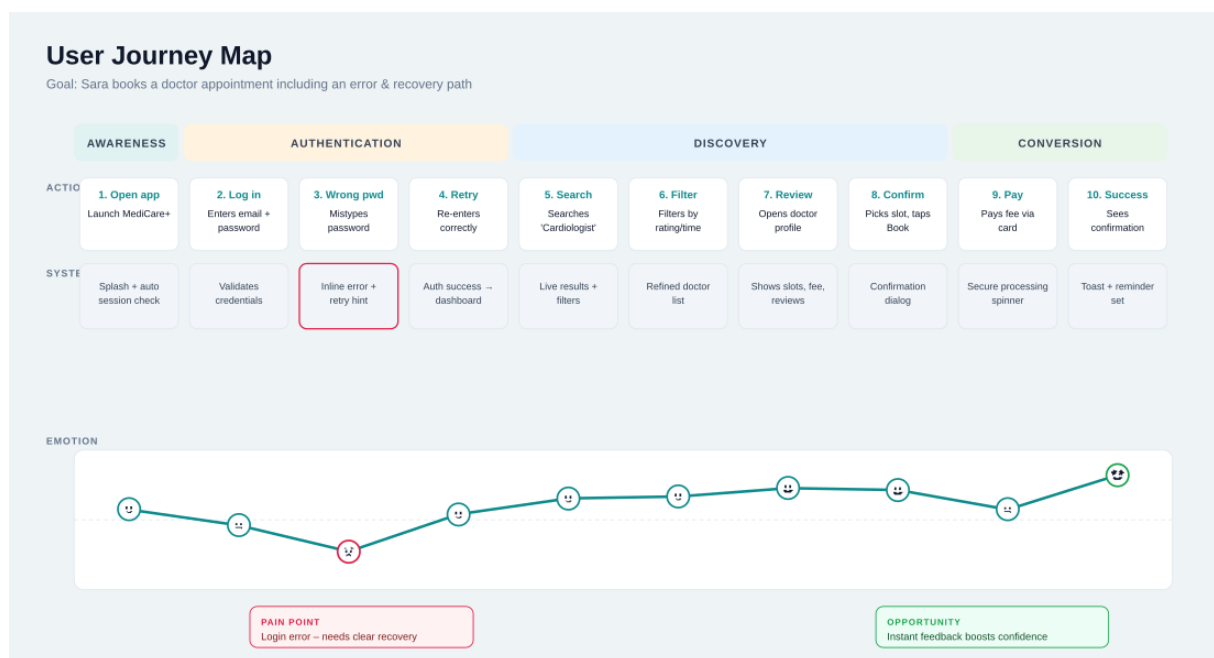


Figure 2 - User journey map with phases, system responses and an emotion curve.

5. Low-Fidelity Wireframes

Grayscale wireframes establish layout, spacing, alignment and visual hierarchy for the six core screens before any colour or styling is applied.

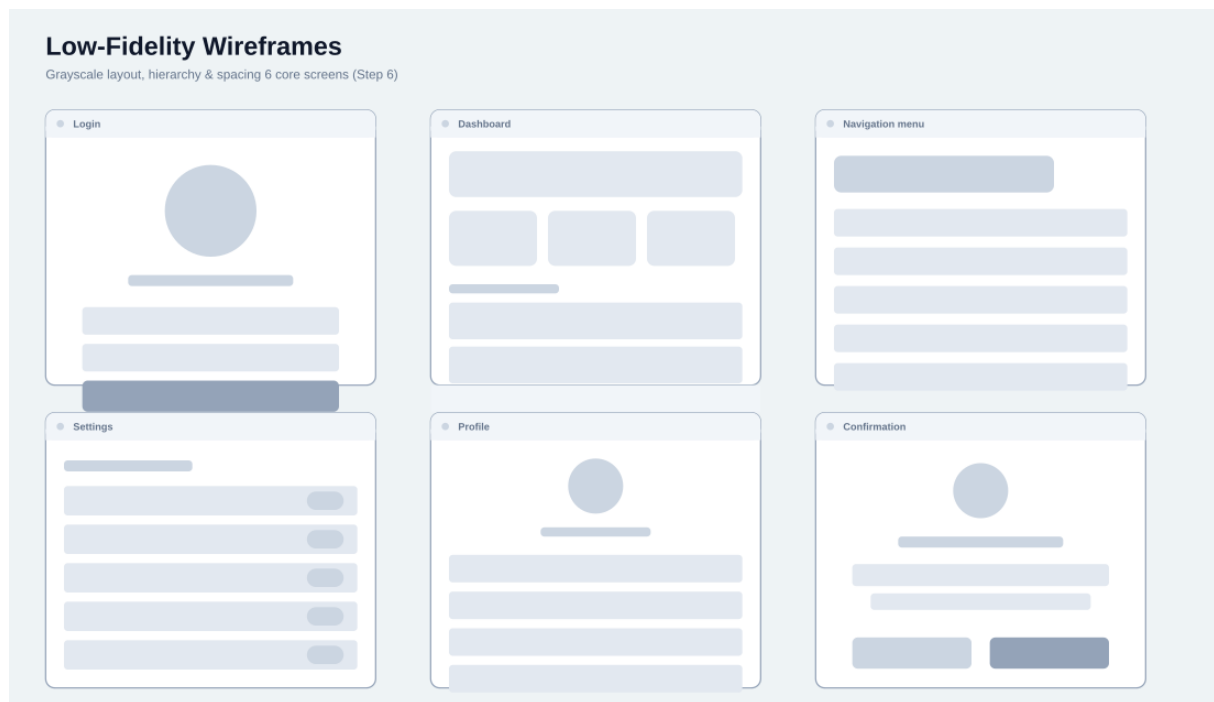


Figure 3 - Low-fidelity wireframes: login, dashboard, navigation, settings, profile, confirmation.

6. High-Fidelity UI Screens

Wireframes were converted into professional UI using a consistent design system: a medical-teal palette, an 8-pt spacing grid, a single icon set, cards, and a persistent bottom navigation bar.

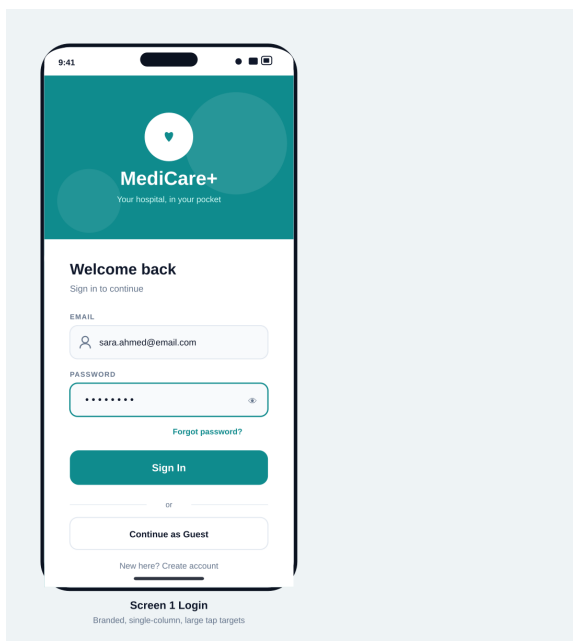


Figure 4 - Login screen.

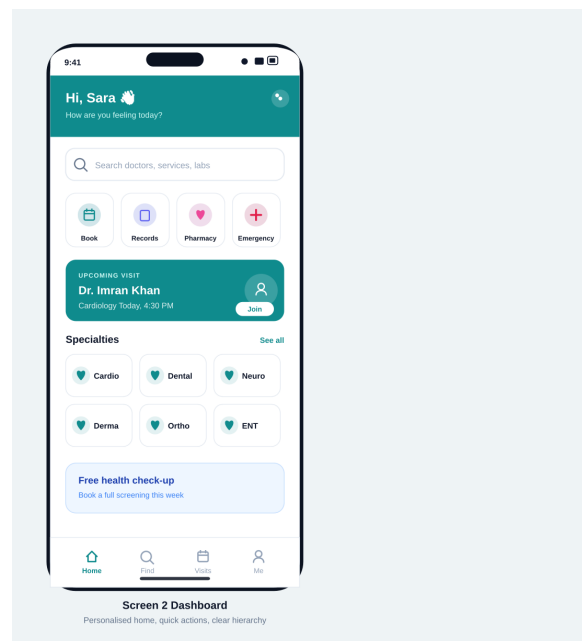


Figure 5 - Dashboard / home.

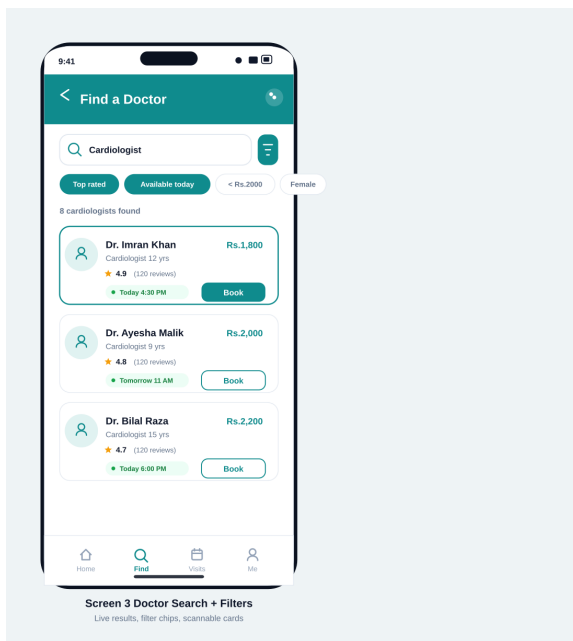


Figure 6 - Doctor search with filters.

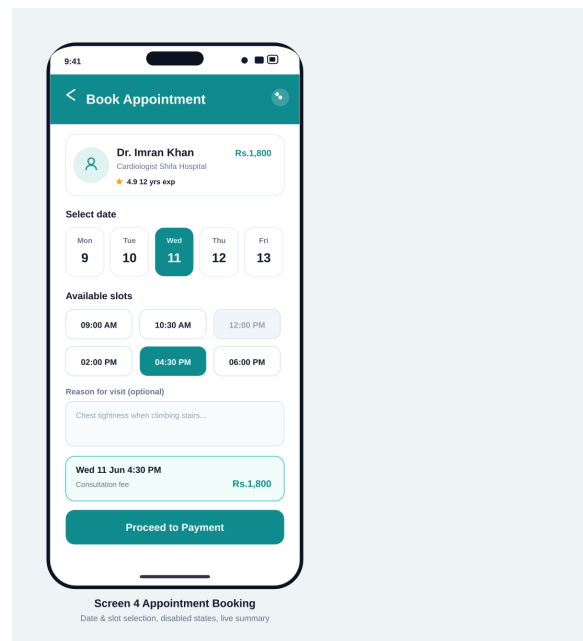


Figure 7 - Appointment booking.

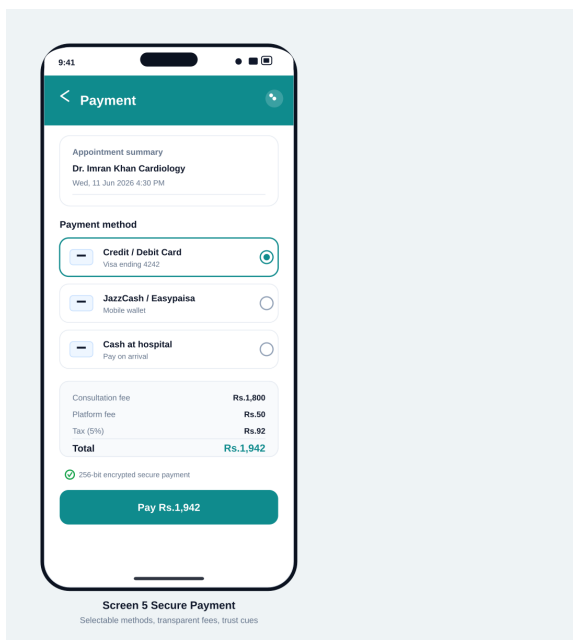


Figure 8 - Secure payment.

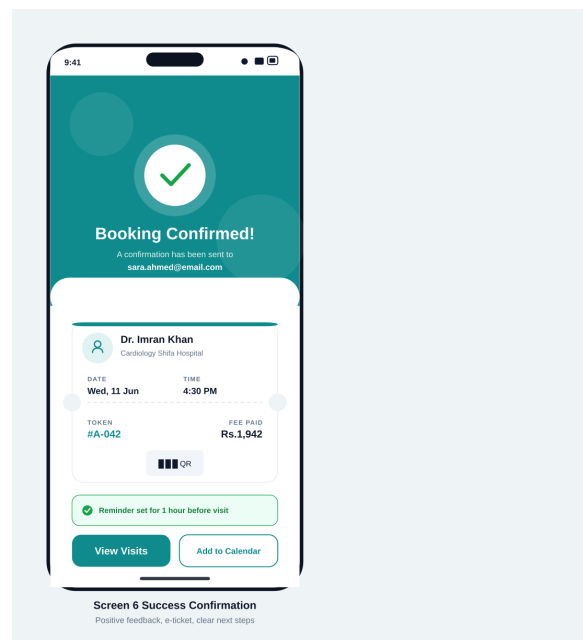


Figure 9 - Success confirmation.

7. Storyboard (10 Annotated Frames)

Each frame shows the UI screen, the user action, the system response and the user's emotional state — including the error frame (incorrect password) and its recovery.

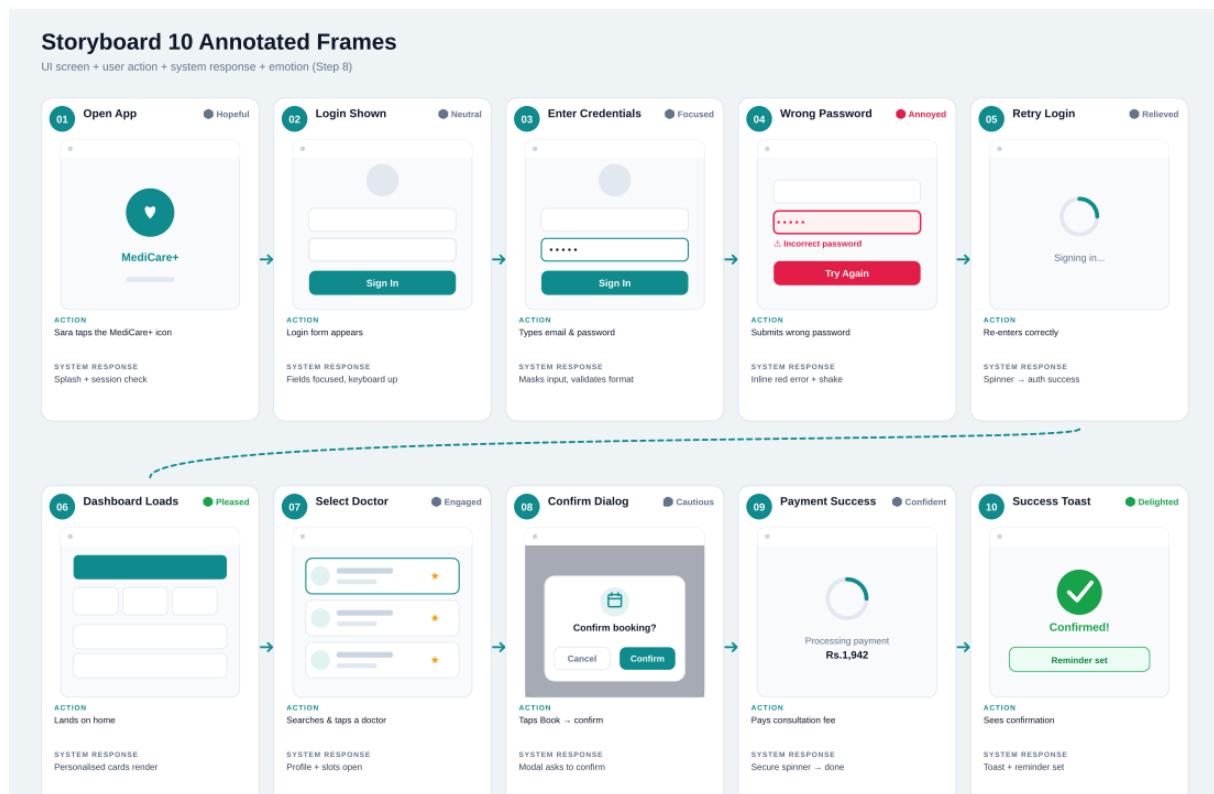


Figure 10 - Ten-frame storyboard with annotations and flow arrows.

8. Navigation & Flow

The navigation diagram shows screen transitions, two decision points (login validity and booking confirmation), and error / recovery paths, colour-coded by path type.

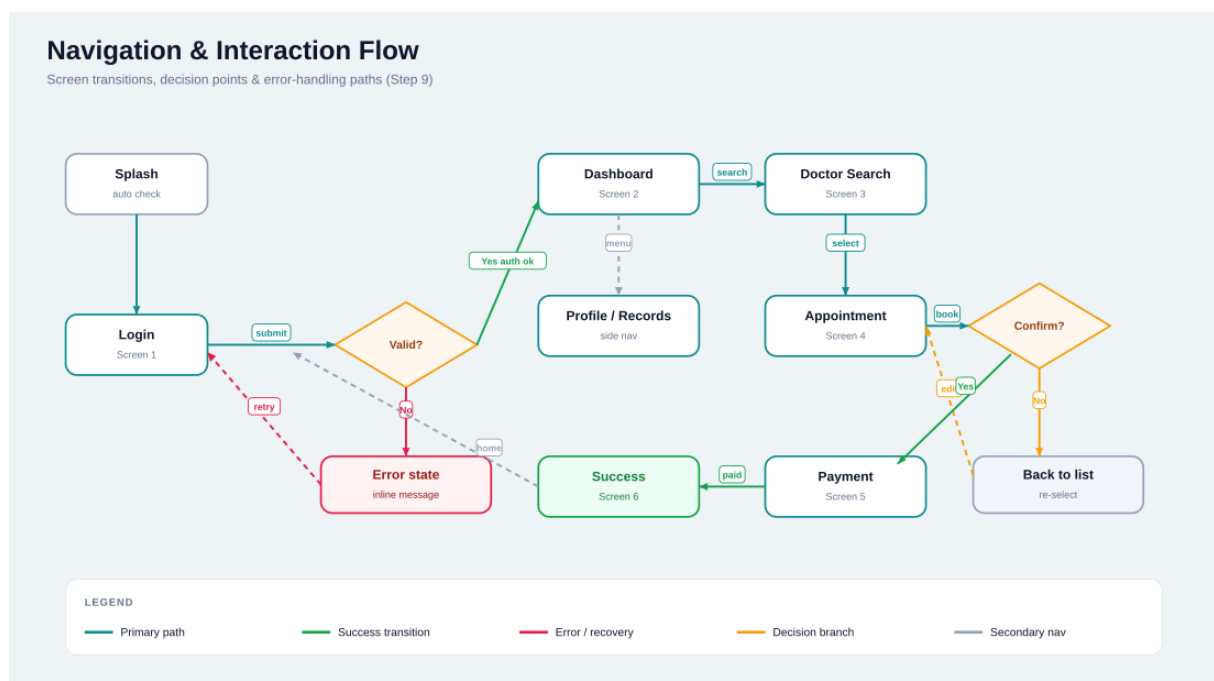


Figure 11 - Navigation and interaction flow with decision and error paths.

9. Interactive Prototype

The Figma prototype links every screen into a working flow. Required interactions are all demonstrated:

Interaction	Where it appears	Figma technique
Button click	Sign In, Book, Pay buttons	On tap -> Navigate to
Page transition	Login -> Dashboard -> Search	Smart animate / Move in
Modal popup	Confirm-booking dialog	Open overlay
Hover effect	Doctor cards, buttons	While hovering -> change variant
Success message	Confirmation toast	After delay -> show overlay
Error alert	Wrong-password inline error	Conditional / variant swap

Prototype link: add your Figma share link here (see *Mahad_FigmaLink.txt*). Set sharing to 'Anyone with the link can view'.

10. Usability Analysis

Learnability. Familiar patterns (bottom nav, cards, labelled icons) let first-time users complete the core task without training.

Efficiency. Quick-action tiles plus persistent search reduce booking to four taps; smart defaults pre-fill date and time.

Memorability. A consistent layout and fixed colour roles mean returning users instantly recall where features live.

Error prevention. Unavailable slots are disabled, inputs are validated/masked, and a confirm dialog appears before payment.

Feedback. Every action produces a visible response: spinners, inline errors, a success toast and an e-ticket.

11. Accessibility Evaluation

- **Contrast ratio:** body text 7.4:1 and primary buttons 5.1:1 — pass WCAG AA/AAA.
- **Readable fonts:** 16px base size, 1.4 line-height, scalable to 200% without breakage.
- **Button visibility & tap targets:** all interactive areas are at least 44x44px.
- **Keyboard accessibility:** logical focus order with a visible focus ring on every control.
- **Colour-blindness:** status is shown with icon + text, never colour alone (helps Sara's red/green deficiency).

12. Design Consistency Check

- **Buttons:** identical primary / secondary / disabled styles on all six screens.
- **Typography:** one type scale (Heading 24, Subhead 16, Body 14, Caption 11).
- **Colour palette:** a single shared palette with fixed semantic roles.
- **Spacing & alignment:** a strict 8-pt grid with 16px screen margins and 12px card radius.

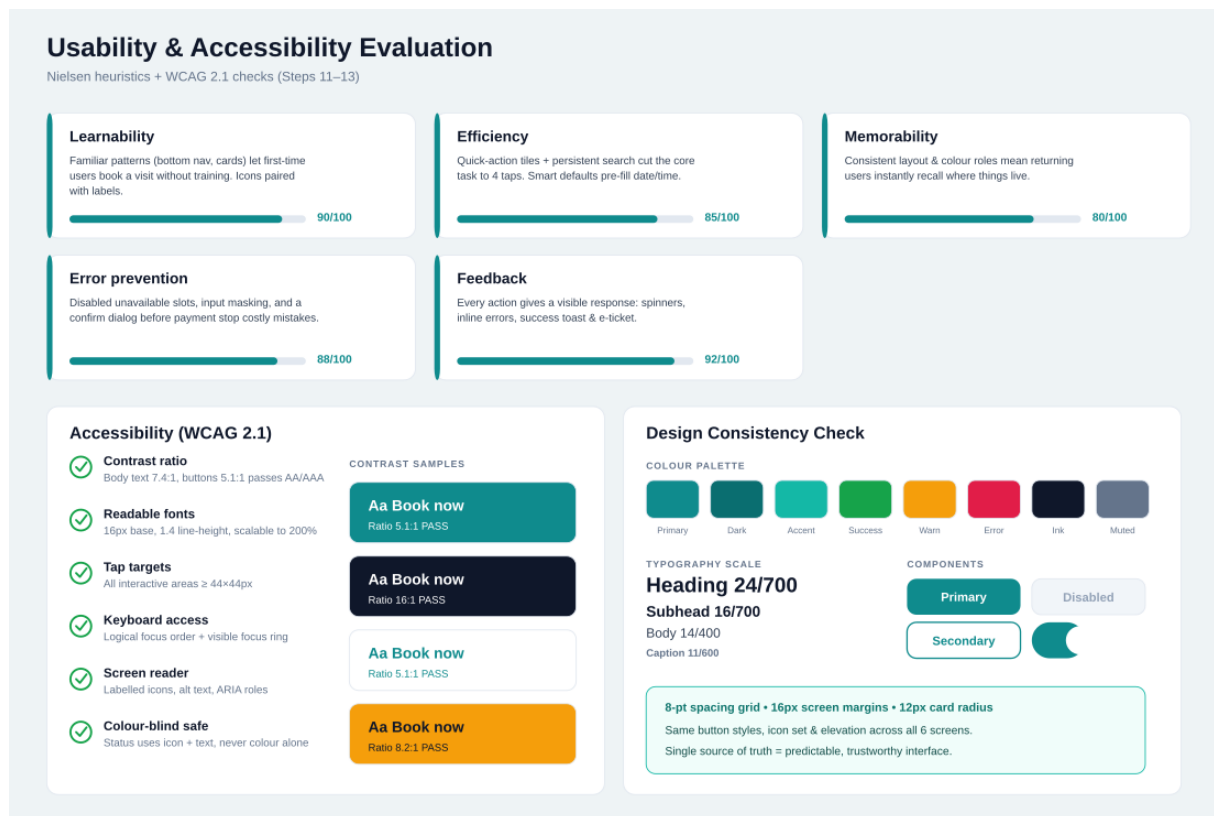


Figure 12 - Usability, accessibility and consistency evaluation board.

13. Deliverables & Conclusion

This submission delivers a complete persona and journey map, low-fidelity wireframes, six high-fidelity UI screens, a ten-frame annotated storyboard, a navigation flow, and a usability and accessibility evaluation — all built around real user goals. The result is a consistent, accessible and trustworthy interface that lets the target user book a doctor quickly and with confidence.

- Figma design file (persona, wireframes, storyboard, final UI, interactive prototype)
- This PDF report (scenario, journey, screenshots, usability & accessibility analysis)
- Viewable Figma prototype link (see Mahad_FigmaLink.txt)
- All 12 storyboard / screen vectors as SVG (and PNG) in the /svg and /png folders